Global Carbon Compliance Architecture: India's CCTS vs. The World (2026–2035) — Structural Mechanics, Price Discovery, and Cross-Border Arbitrage

Kaustav Goswami
Techno India School of Management Studies

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Abstract

This paper presents an institutional-grade market microstructure and econometric evaluation comparing India's Carbon Credit Trading Scheme (CCTS) against six primary global compliance carbon regimes: the European Union Emissions Trading System (EU-ETS), the United Kingdom Emissions Trading Scheme (UK-ETS), the California–Québec Western Climate Initiative (CA-WCI), the New Zealand Emissions Trading Scheme (NZ-ETS), the Korea Emissions Trading Scheme (K-ETS), and the China National ETS over the 2026–2035 compliance horizon. Using a four-layer quantitative simulation engine calibrated to open-access registry data from the European Environment Agency (EEA), the World Bank Carbon Pricing Dashboard, the International Carbon Action Partnership (ICAP), the California Air Resources Board (CARB), and India's Bureau of Energy Efficiency (BEE), we establish three core results. First, we prove via coupled ordinary differential equations (ODEs) that output-adjusted emission intensity baselines decouple GDP growth from absolute emission reductions whenever industrial output growth $g_Y(t)$ exceeds statutory intensity tightening $\gamma_\iota(t)$, generating a cumulative 10-year decoupling wedge of $+2,766.92$ $MtCO_2e$ across India's nine notified CCTS sectors relative to an EU-equivalent 4.3% linear reduction cap. Second, we show that India's statutory environmental compensation cliff under Section 26(1A) of the Energy Conservation (Amendment) Act, 2022 (2× average clearing price) acts as an unbacked financial soft cap without physical allowance retirement. Third, using a $10,000$-path Cholesky-correlated Ornstein–Uhlenbeck jump-diffusion model, we prove that Indian exporters suffer a 94.6%–95.0% *Intensity Recognition Haircut* under Article 9 of the EU Carbon Border Adjustment Mechanism (EU-CBAM), leaving 98.83% of Indian iron and steel border tariffs unhedged and generating a cumulative 2026–2035 export tariff liability of EUR 9,180.65 million under EU-CBAM and GBP 745.89 million under UK-CBAM.

1 Introduction & Global Compliance Landscape

The global compliance carbon landscape entering the 2026–2035 decade is bifurcated into two fundamentally distinct institutional designs. Mature OECD jurisdictions—specifically the EU-ETS (Phase IV), the UK-ETS, the California–Québec WCI Cap-and-Invest Program, the New Zealand ETS, and the Korea ETS (Phase 4)—enforce exogenous, declining *absolute mass caps* (tCO_2e) backed by quantity-based supply adjusters or hard reserve price collars. By contrast, emerging industrial economies led by India's Carbon Credit Trading Scheme (CCTS) and the initial architecture of the China National ETS utilize *output-adjusted emission intensity baselines* (tCO_2e per tonne of physical output).

With the EU Carbon Border Adjustment Mechanism (EU-CBAM) entering its definitive financial surrender period on January 1, 2026, and the UK-CBAM scheduled for implementation in January 2027, the structural divergence between intensity-based domestic compliance and absolute-cap border adjustments creates severe cross-border trade friction. Simultaneously, operationalization of bilateral Internationally Transferred Mitigation Outcomes (ITMOs) under UNFCCC Article 6.2 and the centralized Article 6.4 Paris Agreement Crediting Mechanism (PACM) introduces a sovereign arbitrage boundary governed by Host Party Corresponding Adjustments (CA).

Table 1: Institutional & Containment Comparison Across Seven Global Compliance Hubs (2026 Baseline)

Jurisdiction	**Regime Architecture**	**Unit**	**2026 Cap / Vol.**	**2026 Spot**	**Statutory Stability & Containment Operator $\mathcal{M}(\mathbf{TNAC}_t, P_t)$**
EU-ETS	Absolute Cap ($\alpha_{\text{LRF}} = 4.3\% \to 4.4\%$)	EUA	1,386.0 Mt	EUR 84.50	Market Stability Reserve (MSR): 24% intake above 1,096 Mt (833–1,096 Mt buffer); 100 Mt release below 400 Mt; holdings above prior auction invalidated.
UK-ETS	Absolute Cap ($\alpha_{\text{LRF}} = 4.2\% \to 4.4\%$)	UKA	93.5 Mt	GBP 52.80	Auction Reserve Floor (GBP 22.00) & Cost Containment Mechanism (CCM) triggered at 2.5× trailing 2-year average for 3 consecutive months.
CA-WCI	Absolute Cap ($\alpha_{\text{LRF}} = 4.8\% \to 5.0\%$)	CCA	268.5 Mt	USD 39.40	Auction Floor (USD 25.85, +5% real + CPI), Two-Tier APCR (USD 59.90 / USD 76.85, 15.5 Mt each), Hard Price Ceiling (USD 94.80).
NZ-ETS	Absolute Cap ($\alpha_{\text{LRF}} = 4.6\% \to 4.8\%$)	NZU	24.5 Mt	NZD 64.50	Auction Floor (NZD 64.00) & Two-Tier Cost Containment Reserve (CCR Tier 1 at NZD 184, 2.8 Mt; Tier 2 at NZD 230, 4.9 Mt).
K-ETS	Absolute Cap ($\alpha_{\text{LRF}} = 3.8\% \to 4.2\%$)	KAU	515.0 Mt	KRW 12,400	Phase 4 Korean MSR (K-MSR): 25 Mt reserve pool operating on ±15% circulation corridor and KRW 8,500–28,000 bounds.
China ETS	Intensity → 2028 Abs. Cap	CEA	8,000.0 Mt	CNY 86.50	MEE Benchmark Tightening ($\gamma_\iota = 3.2\%$ in 2026–2027) pivoting to a dual-control absolute cap ($\alpha_{\text{LRF}} = 5.0\%$) in 2028; 5% CCER offset limit.
India CCTS	Output-Adjusted Intensity	CCC	742.5 Mt	INR 980.00	Energy Conservation Act Section 26(1A) Penalty (2.0× average clearing price, unbacked) & CERC exchange collar (INR 650–1,850/t).

2 Mathematical Framing of the Comparative Engine

2.1 Decoupling ODE: Absolute Cap vs. Output-Adjusted Intensity Baseline

Let aggregate industrial output in sector s at time t be $Y_{s,t}$ expanding at instantaneous rate $g_{Y,s}(t) = \dot{Y}_{s,t}/Y_{s,t}$, and let emission intensity be $\iota_{s,t} = E_{s,t}/Y_{s,t}$. Under an *Absolute Cap-and-Trade* regime, aggregate allowance supply S_t^{abs} is fixed exogenously by a statutory Linear Reduction Factor $\alpha_{\text{LRF}} > 0$ relative to base-year cap $\bar{E}_0$:

$$S_t^{\text{abs}} = \bar{E}_0 \left(1 - \alpha_{\text{LRF}} \cdot t\right), \qquad \frac{\partial S_t^{\text{abs}}}{\partial Y_t} = 0. \tag{1}$$

Conversely, under an *Output-Adjusted Intensity Baseline* regime such as India's CCTS, the regulator prescribes a declining target intensity $\iota_{s,t}^* = \iota_{s,0}^* \exp\left(-\int_0^t \gamma_{\iota,s}(u)\, du\right)$. An obligated entity i with physical output $y_{i,t}$ and verified intensity $\iota_{i,t}$ receives Carbon Credit Certificates (CCCs) if $\iota_{i,t} < \iota_{s,t}^*$ and surrenders CCCs if $\iota_{i,t} > \iota_{s,t}^*$:

$$\Delta Q_{i,t}^{\text{CCC}} = y_{i,t}\left(\iota_{s,t}^* - \iota_{i,t}\right) \quad \Longrightarrow \quad S_{s,t}^{\text{int}} = \sum_{i \in s} y_{i,t}\iota_{s,t}^* = Y_{s,t}\iota_{s,t}^*, \qquad \frac{\partial S_{s,t}^{\text{int}}}{\partial Y_{s,t}} = \iota_{s,t}^* > 0. \tag{2}$$

Differentiating $\ln E_{s,t}^{\text{int}} = \ln Y_{s,t} + \ln \iota_{s,t}^*$ with respect to time yields the fundamental coupled ODE system:

$$\frac{dY_{s,t}}{dt} = g_{Y,s}(t) Y_{s,t}, \qquad \frac{d\iota_{s,t}^*}{dt} = -\gamma_{\iota,s}(t)\iota_{s,t}^*, \qquad \frac{\dot{E}_{s,t}^{\text{int}}}{E_{s,t}^{\text{int}}} = g_{Y,s}(t) - \gamma_{\iota,s}(t). \tag{3}$$

Proposition 1 (Macroeconomic Decoupling Condition). *Under 100% market-wide compliance in an output-adjusted intensity regime, absolute sectoral emissions strictly increase ($\dot{E}_{s,t}^{int} > 0$) whenever real output growth exceeds the statutory intensity tightening rate, $g_{Y,s}(t) > \gamma_{\iota,s}(t)$. Instantaneous emissions peaking requires $\gamma_{\iota,s}^*(t) \geq g_{Y,s}(t)$ in continuous time and $\gamma_{\iota,s}^{disc,*} = g_{Y,s}/(1 + g_{Y,s})$ in discrete annual compliance settlement.*

2.2 Piecewise Containment Operators & The Section 26(1A) Penalty Cliff

For each jurisdiction k, the market stability operator $\mathcal{M}_k(\text{TNAC}_t, X_{k,t})$ maps circulation and shadow marginal abatement cost $X_{k,t}$ into effective auction supply adjustments $\Delta S_{k,t}$ and constrained clearing prices $P_{k,t}$. For the EU-ETS MSR, let TNAC_{t-1} denote prior-year Total Number of Allowances in Circulation and S_{t-1}^{auc} denote prior-year auction volume:

$$\Delta S_t^{\text{MSR}} = \begin{cases} -0.24 \cdot \text{TNAC}_{t-1}, & \text{if } \text{TNAC}_{t-1} > 1,096 \text{ Mt}, \\ -\left(\text{TNAC}_{t-1} - 833 \text{ Mt}\right), & \text{if } 833 \text{ Mt} < \text{TNAC}_{t-1} \leq 1,096 \text{ Mt}, \\ 0, & \text{if } 400 \text{ Mt} \leq \text{TNAC}_{t-1} \leq 833 \text{ Mt}, \\ +\min\left(100 \text{ Mt},\ M_{t-1}^{\text{MSR}}\right), & \text{if } \text{TNAC}_{t-1} < 400 \text{ Mt}, \end{cases} \tag{4}$$

subject to permanent invalidation $I_t^{\text{MSR}} = \max\left(0,\ M_t^{\text{pre}} - S_{t-1}^{\text{auc}}\right)$. In contrast, under Section 26(1A) of India's Energy Conservation (Amendment) Act, 2022, an obligated entity failing to surrender CCCs pays an environmental compensation penalty equal to twice the average clearing price across the compliance cycle, $P_t^{\text{pen}} = 2\bar{P}_{t,\text{CCC}}$, intersected with CERC exchange bands $\left[P_t^{\text{floor}}, P_t^{\text{forb}}\right]$:

$$P_{\text{IND},t} = \min\left(\min\left(P_t^{\text{forb}},\ 2\bar{P}_{t,\text{CCC}}\right),\ \max\left(P_t^{\text{floor}},\ X_{\text{IND},t}\right)\right). \tag{5}$$

Because Section 26(1A) compensation is a financial levy rather than a physical allowance reserve drawn from beneath a hard cap, any excursion of industrial marginal abatement cost $X_{\text{IND},t} > \min\left(P_t^{\text{forb}}, 2\bar{P}_{t,\text{CCC}}\right)$ induces rational financial buy-out without backed physical abatement.

2.3 Correlated OU Jump-Diffusion, CBAM Haircut & Article 6.4 Gross-Up

Across the seven compliance hubs $k \in \{1, \dots, 7\}$, unconstrained shadow marginal abatement costs evolve according to a correlated mean-reverting Constant Elasticity of Variance (CEV) jump-diffusion system:

$$dX_{k,t} = \kappa_k \left[\theta_k \left(D_{k,t}^{\text{net}}, S_{k,t}^{\text{eff}}\right) - X_{k,t}\right] dt + \sigma_k X_{k,t}^{\beta_k} dW_{k,t} + X_{k,t-} J_{k,t} dN_{k,t}, \qquad d\mathbf{W}_t = \mathbf{L}\, d\mathbf{Z}_t, \tag{6}$$

where $\mathbf{\Sigma}_{\text{corr}} = \mathbf{L}\mathbf{L}^\top$ is the Cholesky factorization of the 7×7 cross-hub correlation matrix, $d\mathbf{Z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_7 dt)$, and $dN_{k,t} \sim \text{Poisson}(\lambda_k dt)$ captures regulatory and fuel-switching shocks.

Under Articles 9 and 31 of the EU-CBAM Regulation (EU) 2023/956, the net border tariff per tonne of exported good in sector s from jurisdiction j to destination $d \in \{\text{EU, UK}\}$ at time t is:

$$\tau_{j \to d,s,t}^{\text{CBAM}} = \max\left(0,\ \left[e_{j,s,t}^{\text{actual}} - \Phi_{d,t}^{\text{CBAM}} B_{d,s}^{\text{bench}}\right] P_{d,t}^{\text{ETS}} - C_{j,s,t}^{\text{paid,FX}}\right). \tag{7}$$

Under India's intensity-based CCTS, an exporting plant only pays for CCCs on the marginal intensity gap $\max(0, \iota_{s,t} - \iota_{s,t}^*)$ rather than total embedded emissions $e_{j,s,t}^{\text{actual}} = \iota_{s,t}$. Defining the *Intensity Recognition Haircut* $\mathcal{H}_{s,t}^{\text{int}}$ relative to a full-scope domestic carbon price $C_{s,t}^{\text{full,FX}} = \iota_{s,t} P_{\text{IND},t}^{\text{CCC}} \text{FX}_t$ yields:

$$C_{\text{IND},s,t}^{\text{paid,FX}} = \max\left(0, \iota_{s,t} - \iota_{s,t}^*\right) P_{\text{IND},t}^{\text{CCC}} \text{FX}_t \quad \Longrightarrow \quad \mathcal{H}_{s,t}^{\text{int}} = 1 - \frac{C_{\text{IND},s,t}^{\text{paid,FX}}}{C_{s,t}^{\text{full,FX}}} = \frac{\iota_{s,t}^*}{\iota_{s,t}}. \tag{8}$$

Finally, for bilateral ITMO transfers under Article 6.2 and Article 6.4 PACM, incorporating the mandatory 5% Adaptation Fund Share of Proceeds ($\lambda_{\text{SOP}} = 0.05$), 2% Overall Mitigation in Global Emissions cancellation ($\lambda_{\text{OMGE}} = 0.02$), Supervisory Body fee c_{fee}, and Host Country NDC opportunity cost $\Omega_{\text{NDC},t}^{\text{CA}}$ yields the sovereign arbitrage condition:

$$P_{\text{buyer},t} \geq P_{\text{min,ITMO},t} = \frac{\text{MAC}_{\text{host},t} + \Omega_{\text{NDC},t}^{\text{CA}}}{1 - \lambda_{\text{SOP}} - \lambda_{\text{OMGE}}} + c_{\text{fee}} = 1.07527 \left(\text{MAC}_{\text{host},t} + \Omega_{\text{NDC},t}^{\text{CA}}\right) + c_{\text{fee}}. \tag{9}$$

3 Empirical Findings & Multi-Hub Benchmarking

3.1 India 9-Sector Decoupling Wedge & China's 2028 Dual-Control Pivot

Integrating Eq. (3) across all nine notified BEE CCTS sectors demonstrates that under India's industrial expansion rates ($g_{Y,s} \in [4.2\%, 7.4\%]$), aggregate compliance emissions rise by +22.75% from 865.11 MtCO_2e in 2026 to 1,061.93 MtCO_2e in 2035, creating a cumulative 10-year decoupling wedge of +2,766.92 MtCO_2e over a 4.3% linear reduction cap (Table 2). Only Fertilizers ($g_Y = 4.5\% < \gamma_{\text{P2}} = 5.0\%$) and Petroleum Refining ($g_Y = 4.2\% < \gamma_{\text{P2}} = 4.8\%$) achieve absolute peaking before 2035. By contrast, China's scheduled 2028 dual-control pivot from an intensity baseline ($g_Y = 4.5\%, \gamma_\iota = 3.2\%$) to an absolute cap ($\alpha_{\text{LRF}} = 5.0\%$) reduces 2035 covered emissions from 8,992.96 Mt to 4,862.81 Mt, avoiding 18,525.05 MtCO_2e cumulatively over 2026–2035.

Key Structural Finding: Because allowance creation under an intensity baseline is endogenous to industrial output ($\partial S_t^{\text{int}} / \partial Y_t = \iota_t^* > 0$), India's weighted average Phase 2 tightening rate (4.79%/yr) falls 182 bps short of the 6.61%/yr threshold required merely to stabilize absolute industrial emissions, and 718 bps short of the 11.97%/yr rate required to match a 4.3%/yr linear cap reduction.

Table 2: India CCTS 9-Sector ODE Decoupling Trajectory & Peaking Thresholds (2026–2035)

BEE CCTS Sector	CBAM Exp.	2026 Base (Mt)	2035 Int. (Mt)	2035 Cap (Mt)	10-Yr Wedge ($MtCO_2e$)	$g_{Y,s}$ (%/yr)	γ_{P2} (%/yr)	Peak $\gamma^*_{t,s}$ (%/yr)	Cap-Eq. $\bar{\gamma}^*_{t,s}$ (%/yr)
Iron & Steel	Yes	337.44	437.64	206.85	+1,193.19	7.20%	4.80%	7.20%	12.56%
Cement	Yes	253.15	324.08	155.18	+871.18	6.80%	4.50%	6.80%	12.16%
Aluminum	Yes	65.12	74.91	39.92	+186.15	6.50%	5.50%	6.50%	11.86%
Fertilizers	Yes	54.18	54.13	33.21	+113.01	4.50%	5.00%	4.50%	9.86%
Chlor-Alkali	No	6.14	6.95	3.76	+16.89	5.80%	4.90%	5.80%	11.16%
Pulp & Paper	No	24.15	26.16	14.80	+60.56	5.50%	5.10%	5.50%	10.86%
Petrochemicals	No	41.89	52.99	25.68	+142.13	7.40%	5.30%	7.40%	12.76%
Petroleum Refining	No	61.92	61.30	37.96	+126.34	4.20%	4.80%	4.20%	9.56%
Textiles	No	21.12	23.77	12.95	+57.47	6.20%	5.40%	6.20%	11.56%
Total / Weighted	**4 Sect.**	**865.11**	**1,061.93**	**530.31**	**+2,766.92**	**6.61%**	**4.79%**	**6.61%**	**11.97%**

3.2 10,000-Path Correlated Price Discovery & Carbon-VaR Distributions

Table 3 summarizes the 10,000-path Cholesky-correlated OU jump-diffusion price distributions across 2026, 2030, and 2035. While Indian CCC median clearing prices rise from INR 1,042.59/t (EUR 11.38/t) in 2026 to INR 2,129.37/t (EUR 23.25/t) in 2035, the median EU-ETS vs. India CCTS basis spread widens from EUR 74.62/t in 2026 to EUR 88.13/t in 2030 and EUR 108.95/t in 2035 (with the 99% Carbon-VaR basis spread reaching EUR 150.18/t). By 2035, Indian industrial marginal abatement costs breach the Section 26(1A) penalty cliff or CERC forbearance ceiling in 11.61% of simulated paths.

Table 3: 10,000-Path Correlated OU Jump-Diffusion Price Discovery & Carbon-VaR Summary (2026, 2030, 2035)

Hub	Curr.	2026 Med.	2026 VaR_{95}	2030 Med.	2030 VaR_{95}	2035 Med.	2035 VaR_{99}	2035 EUR Med. / Spread
EU-ETS	EUR	85.99	101.03	103.94	125.66	132.28	173.75	EUR 132.28 (—)
UK-ETS	GBP	54.21	65.43	65.10	81.72	81.57	114.88	EUR 96.54 (EUR 35.72)
CA-WCI	USD	40.74	44.90	53.36	59.55	75.10	88.12	EUR 68.90 (EUR 63.36)
NZ-ETS	NZD	66.99	77.03	82.58	96.91	107.74	138.02	EUR 60.29 (EUR 71.97)
K-ETS	KRW	12,959	14,968	16,312	19,423	21,571	27,451	EUR 14.64 (EUR 117.63)
China ETS	CNY	89.55	97.02	119.99	131.26	172.58	193.70	EUR 22.17 (EUR 110.08)
India CCTS	INR	1,042.59	1,190.59	1,439.13	1,679.77	2,129.37	2,557.22	EUR 23.25 (EUR 108.95)

3.3 EU/UK-CBAM Intensity Recognition Haircut & 10-Year Export Liabilities

Table 4 reports the empirical evaluation of Eq. (7)–(8) across the four CBAM-exposed Indian sectors. Because Indian facilities pay domestic CCC prices only on the marginal intensity gap ($\iota_{s,t} - \iota^*_{s,t}$), their Article 9 carbon deduction suffers a mean *Intensity Recognition Haircut* of 94.62% to 95.03%. Across iron and steel, aluminum, and fertilizers, 97.86% to 98.83% of the gross EU-CBAM border tariff remains unhedged, generating a cumulative 10-year liability of EUR 9,180.65 million under EU-CBAM and GBP 745.89 million under UK-CBAM.

Table 4: India 4-Sector EU-CBAM & UK-CBAM Tariff Exposure and Article 9 Intensity Haircut (2026–2035)

Sector	2026 EU (EUR/t)	2030 EU (EUR/t)	2035 EU (EUR/t)	2035 VaR_{95} (EUR/t)	Haircut (%)	Unhedged Share (%)	10-Yr EU (EUR M)	10-Yr UK (GBP M)
Iron & Steel	90.69	134.52	210.83	256.39	94.81%	98.83%	7,924.61	620.23
Aluminum	61.26	104.33	181.65	220.91	94.62%	98.59%	947.64	84.13
Fertilizers	20.75	69.95	156.52	190.34	94.83%	97.86%	207.59	27.80
Cement	0.00	18.39	57.66	70.11	95.03%	67.53%	100.81	13.73
Total / Mean	—	—	—	—	**94.82%**	**98.43%**	**9,180.65**	**745.89**

4 Strategic Policy Implications for India

Our quantitative findings yield three actionable policy imperatives for India's Ministry of Power, the Bureau of Energy Efficiency (BEE), and the National Steering Committee for Indian Carbon Market (NSC-ICM):

1. **Dual-Track Export Equivalence & Domestic Carbon Fee Retention:** Rather than allowing EUR 9.18 billion in CBAM certificates and GBP 745.89 million in UK-CBAM levies to be transferred to European and British treasuries over 2026–2035, India should introduce an optional *Full-Scope Export Vintage Tier* within CCTS for EU/UK-bound consignments in iron and steel, aluminum, fertilizers, and clinker, converting the unhedged border tariff into domestic industrial decarbonization revenue.

2. **Replacing the Unbacked Section 26(1A) Penalty Cliff with a Physical Containment Reserve:** To ensure environmental integrity and enable future registry linkage, Section 26(1A) of the Energy Conservation Act should be amended so that forbearance ceiling interventions inject allowances from a finite, pre-escrowed *Indian Cost Containment Reserve (I-CCR)* rather than permitting unbacked financial fee settlement.

3. **Sovereign Article 6.2/6.4 ITMO Reservation Floor ($\Omega_{\mathbf{NDC}}^{\mathbf{CA},*}$):** As shown in Eq. (9), with a 1.07527× PACM gross-up multiplier and a 2030 minimum viable ITMO offer price of USD 31.37/tCO_2e, India enjoys positive bilateral arbitrage spreads against the EU-ETS (+USD 81.92/t), UK-ETS (+USD 52.60/t), CA-WCI (+USD 21.99/t), and NZ-ETS (+USD 19.00/t), while K-ETS (−USD 19.30/t) remains sub-threshold. India should restrict Corresponding Adjustment authorizations strictly to high-MAC technologies (green hydrogen direct reduced iron, industrial CCUS, and offshore wind) priced above USD 45/tCO_2e so low-cost abatement (<USD 25/t) is retained for domestic CCTS compliance.

5 Conclusion

By uniting continuous-time decoupling ODEs, piecewise statutory containment operators, and a 10,000-path Cholesky-correlated jump-diffusion engine, this study demonstrates that output-adjusted emission intensity regimes cannot achieve absolute emission reductions or shield trade-exposed industries from border carbon adjustments when real industrial growth exceeds intensity tightening rates. A staged transition toward a dual-control absolute cap—paired with a physical Market Stability Reserve and a sovereign Article 6 reservation price—is essential to align India's industrial expansion with global carbon market equivalence by 2035.

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